

DAY 1 · MODULE 5 · 35 MIN

The Agent Stack

The mental model we use

The five-layer stack

- Model — the deployed model
- Runtime — where the agent runs (Prompt agent, Hosted agent, or your own code calling the Responses API)
- Actions — how it does things (tools, Toolbox, MCP)
- Knowledge — how it knows things (Foundry IQ, RAG)
- Ops — identity, tracing, evaluation, cost, deployment

Keep this five-word list in your head all week.

Layer 1 — Model

Where it lives: your Foundry project, as a model deployment.

- Which model? (frontier vs. efficient, cost vs. capability)
- What deployment capacity? (tokens per minute)
- Which region?

Day 5 covers model routing — small model first, escalate on low confidence.

Layer 2 — Runtime

Where the agent code and state live. Two hosting families:

- Foundry-hosted · Prompt agent — portal- or SDK-authored, no code, Foundry runs it
- Foundry-hosted · Hosted agent — your code, packaged as a container, Foundry runs the container
- Self-hosted — your process runs your code and calls Foundry for models and tools

Module 6 is entirely about this choice. Each family has a sweet spot.

Layer 3 — Actions

How the agent affects the world.

- Function tools — Python or C# functions you write, decorated for MAF
- Foundry Toolbox — curated tools exposed via an MCP endpoint (Bing, Fabric, SharePoint, code interpreter, ...)
- MCP servers — the open protocol; MAF can consume any MCP server and you can author your own
- Skills — portable packages of instructions + reference material + optional scripts, with progressive disclosure

Day 2: function tools and Toolbox. Day 3: MCP end-to-end. Skills: awareness only in this workshop — see the dedicated slide below.

Layer 4 — Knowledge

How the agent grounds its answers.

- Foundry IQ — enterprise knowledge / grounding layer; unified retrieval across AI Search, SharePoint, OneLake / Fabric
- Custom RAG — AI Search or a vector store you drive yourself, when you need control IQ doesn't yet give you

Day 2 deep dive: both, plus when to prefer one over the other.

Layer 5 — Ops

The unglamorous layer that makes agents production-worthy.

- Identity — Entra, managed identity, RBAC
- Tracing — OTEL spans of every model call, tool invocation, and decision; view in the Foundry portal or ship to App Insights
- Evaluation — retrieval, single-agent, multi-agent, continuous
- Cost & latency — model tiers, caching, batching, routing
- Deployment — Container Apps, Functions, AKS

Day 5 owns most of this — but evaluation is threaded through every day.

Skills — packaging expertise for reuse

	Tool	Skill
What it provides	A single callable action	Instructions + reference material + optional scripts
How agent uses it	Calls it when it needs to act	Loads it when task matches; reads instructions; may call scripts
Context cost	Tool schema always in the prompt	Only name + description (~100 tokens) upfront; full content on demand
Portability	Tied to the agent that registers it	Self-contained package any compatible agent can discover
Best for	Individual actions (query a DB, send an email)	Domain expertise (expense policies, code review guidelines)

Rule of thumb: tools are verbs (search, book, validate). Skills are expertise (travel booking knowledge, expense policy knowledge). Agents use tools to act and skills to know how to act.

Where each day lives on the stack

Day	Model	Runtime	Actions	Knowledge	Ops
1	✓	✓ (both flavors)	tease	tease	—
2	✓	✓	deep	deep	eval
3	✓	deep	deep (MCP)	✓	eval
4	✓	multi-agent	✓	✓	eval anchor
5	routing	✓	✓	✓	deep

Why this matters

- Diagnose problems by layer — 'the agent isn't grounded' is a Knowledge problem, not a Model problem
- Estimate cost and risk — adding an MCP server is an Actions + Ops concern, not a model concern
- Communicate cleanly with stakeholders — product, ops, security all live at different layers

Bring this back to your team as a shared vocabulary.

Takeaways

- Five layers: Model, Runtime, Actions, Knowledge, Ops.
- Every day of the workshop, and every real agent you build, fits into these.
- When something breaks or costs too much, ask which layer first.

Next: Module 6 zooms into Layer 2 (Runtime) — the two hosting families for Agent Framework agents.